

# **ARDUINO AS ISP For Programming 89s52 (8051)**

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step1) download python 3.14.7 <https://www.python.org/downloads>

step2) in cmd type pip install pyserial

Step3) generate a hex file (use any compiler ) to compile c file to hex file

We used uvision keil software

step4)save into a easily accessible folder in ur system

Step 5) save at89s52\_programmer.py file to folder

Step 6) open cmd change directory to your folder

Step 7)connect the arduino and 82s59 as per pin

Step 8) upload arduino code given

Step 9) in cmd type python at89s52\_programmer.py ports  
(this will check the ports connection)

```
C:\Users\ASUS\Desktop>python at89s52_programmer.py ports
COM7 - Standard Serial over Bluetooth link (COM7)
COM3 - Standard Serial over Bluetooth link (COM3)
COM6 - USB-SERIAL CH340 (COM6)
COM8 - Standard Serial over Bluetooth link (COM8)
COM4 - Standard Serial over Bluetooth link (COM4)

C:\Users\ASUS\Desktop>
```

Step 10) in cmd type python at89s52\_programmer.py identify -p "your com"

(eg:python at89s52\_programmer.py identify -p COM6)

This will identify 8051 ic

```
C:\Users\ASUS\Desktop>python at89s52_programmer.py identify -p COM6
Signature bytes: 0x1E 0x06 0x52
Manufacturer byte 0x1E matches Atmel/Microchip.

C:\Users\ASUS\Desktop>
```

Step 11) in cmd type python at89s52\_programmer.py upload -p 'your com' -f  
'your\_hexfile'

eg:python at89s52\_programmer.py upload -p COM6 -f blink.hex

```
D:\arduinoisp>dir *.hex
Volume in drive D is New Volume
Volume Serial Number is E866-38C4

Directory of D:\arduinoisp

17-09-2026  10.31 PM                263 lol.hex
             1 File(s)                263 bytes
             0 Dir(s)  208,072,015,872 bytes free

D:\arduinoisp>python at89s52_programmer.py upload -p COM6 -f lol.hex
Erasing chip before upload...
Uploading lol.hex ...
Writing: 100% (73/73)
Wrote 73 bytes.
Verifying...
Verifying: 100% (73/73)
Verify OK.

D:\arduinoisp>
```

This will upload the code and show verify ok if uploaded successfully

step12)if it shows error of not finding hex file use - dir \*.hex

```
C:\Users\ASUS\Desktop>python at89s52_programmer.py upload -p COM6 -f lol.hex
Erasing chip before upload...
Uploading lol.hex ...
Traceback (most recent call last):
  File "C:\Users\ASUS\Desktop\at89s52_programmer.py", line 401, in <module>
    main()
    ^^^^^^^
  File "C:\Users\ASUS\Desktop\at89s52_programmer.py", line 397, in main
    args.func(args)
    ^^^^^^^^^^^^^^
  File "C:\Users\ASUS\Desktop\at89s52_programmer.py", line 318, in cmd_upload
    count = prog.upload(args.file)
  File "C:\Users\ASUS\Desktop\at89s52_programmer.py", line 221, in upload
    mem = parse_intel_hex(hex_path)
  File "C:\Users\ASUS\Desktop\at89s52_programmer.py", line 57, in parse_intel_hex
    with open(path, "r") as f:
         ^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'lol.hex'

C:\Users\ASUS\Desktop>dir *.hex
Volume in drive C is OS
Volume Serial Number is CEA8-E5F2

Directory of C:\Users\ASUS\Desktop

File Not Found

C:\Users\ASUS\Desktop>
```

If above step are done correctly it will give a verify ok message

```
D:\arduinoisp>dir *.hex
Volume in drive D is New Volume
Volume Serial Number is E866-38C4

Directory of D:\arduinoisp

17-09-2026  10.31 PM                263 lol.hex
              1 File(s)              263 bytes
              0 Dir(s) 208,072,015,872 bytes free
```

```
D:\arduinoisp>python at89s52_programmer.py upload -p COM6 -f lol.hex
Erasing chip before upload...
Uploading lol.hex ...
Writing: 100% (73/73)
Wrote 73 bytes.
Verifying...
Verifying: 100% (73/73)
Verify OK.

D:\arduinoisp>
```

Credits to  
Devanarayanan CR  
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The files will be provided with this guide